

Variants of seeded region growing

Minjie Fan, Thomas C.M. Lee

Department of Statistics, University of California at Davis, One Shields Avenue, Davis, CA 95616, USA
 E-mail: tcmlee@ucdavis.edu

Abstract: Seeded region growing (SRG) is a fast, effective and robust method for image segmentation. It begins with placing a set of seeds in the image to be segmented, where each seed could be a single pixel or a set of connected pixels. Then SRG grows these seeds into regions by successively adding neighbouring pixels to them. It finishes when all pixels in the image are assigned to one (and only one) region. The growing strategy of SRG implicitly assumes that pixels from the same region share the same greyvalue. As a first contribution, this study develops new modifications to SRG so that this constant greyvalue assumption is relaxed. Since the growing strategy of SRG does not impose any constraints or restrictions on the shapes of the growing regions, quite often SRG would produce very rough segmentation boundaries even the true boundaries were smooth. The second contribution of this study is the proposal of a stabilised SRG that encourages smoother boundaries and prevents the so-called leakage problem. All these new variants are conceptually simple and easy to implement. They are tested with simulated and real images, and are shown to have better performances than the original SRG.

1 Introduction

Image segmentation aims to locate the boundaries of objects contained in images. The seeded region growing (SRG) method proposed by [1] is a conceptually simple and yet an effective and robust technique for performing segmentation. Since its inception, it has been widely applied by researchers and practitioners to produce successful segmentation results for different real life problems; for example see [2–4]. In addition, various extensions and modifications have been proposed to improve the performance of SRG; for example see [5–7].

The main goal of this paper is to introduce three new variants of SRG to further improve its performance under the following two scenarios. Note that an implicit assumption behind SRG is that all pixels within the same region share the same greyvalue. In other words, all regions in the image are of constant greyvalues, and can be modelled as horizontal planes. The first scenario this paper considers is when this constant greyvalue assumption is violated. To be more specific, the first contribution of this paper is the development of two SRG variants for which the regions are modelled as linear planes and quadratic surfaces, respectively. To the best of the authors' knowledge, there is no existing SRG based method that is designed to cope with linear and quadratic regions.

The second scenario happens when the signal-to-noise ratio of the image is low, or when the boundary between a region and its background is blurry. In this case the so-called leakage problem could occur, which would in turn produce poor segmentation results. As a second contribution, this paper proposes a SRG variant to circumvent this issue by encouraging smoother segmented region boundaries. It is noted that various SRG

modifications have been proposed to produce smoother boundaries [e.g. 6–8], but a distinct advantage of the current proposal is that it can be easily paired with other variants developed in this paper.

The rest of this paper is organised as follows. In Section 2 the original SRG of [1] will first be reviewed, and then it will be modified for enhanced performances. All the new SRG variants will be empirically accessed through a sequence of images in Section 3. In general these new variants improve the results from the original SRG. Concluding remarks are given in Section 4.

2 Variants of SRG

2.1 Original SRG

The original SRG [1] begins segmenting an image from a set of seeds. Each of these seeds could be a single pixel or a group of pixels, and they can be specified manually by a human operator or automatically by pre-processing steps [e.g. 4]. These seeds are then allowed to grow and form regions in a manner to be described below. Finally the whole procedure finishes when the image is fully partitioned by all the growing regions. Object boundaries are obtained as the boundaries of the final regions.

A more precise description of the original SRG is as follows. Suppose initially there are N seeds, and they are collected in N sets, $A_1, A_2, \dots, A_N$. Let T be the set of all pixels which are adjacent to at least one of the pixels in A_i 's

$$T = \left\{ x \notin \bigcup_{i=1}^N A_i \mid \text{nbr}(x) \cap \bigcup_{i=1}^N A_i \neq \emptyset \right\}$$

where $\text{nbr}(x)$ is the set of all immediate neighbours of pixel x . At each time step, the growing process selects one pixel x from T and adds it to one of its neighbouring regions; A_i is a neighbouring region of x if $\text{nbr}(x) \cap A_i \neq \emptyset$. This pixel is chosen according to some homogeneity criterion. In [1] the criterion is to choose the pixel whose greyvalue is closest to the average greyvalue of one of its neighbouring growing regions, say A_p where $1 \leq p \leq N$. This pixel will then be added to that region, and both T and A_p will be updated. This process continues until all the pixels in the image are allocated to one and only one of the growing regions. The criterion of [1] ensures that the grey values for each final region are as homogeneous as possible under the connectivity constraint.

The following notation will prove to be useful in subsequent sections. Denote the pixels of the growing region R under consideration as $R = \{r_1, r_2, \dots, r_n\}$; that is, there are n pixels. Let $g(x)$ be the greyvalue of pixel x . With this notation, the original SRG is to choose the pair of (x, R) such that the following criterion is minimised

$$\delta_O(x, R) = \left| g(x) - \frac{1}{n} \sum_{i=1}^n g(r_i) \right| = |g(x) - \bar{r}|$$

where $\bar{r} = (1/n) \sum_{i=1}^n g(r_i)$ is the averaged greyvalue of R . In below modifications to $\delta_O(x, R)$ are presented to improve the performance of the original SRG.

2.2 Linear SRG

As hinted before, the use of $\delta_O(x, R)$ implicitly assumes that the greyvalues of any region in the image can be well approximated by a single constant value. This subsection relaxes this assumption by modelling the greyvalues with a linear plane (i.e. a plane with a non-zero slope).

Let the numbers of rows and columns of an image be n_r and n_c , respectively. Then the coordinate of a pixel located at the i th row and j th column is $(i/n_r, j/n_c)$. The greyvalue of a pixel is modelled as $a_1(i/n_r) + a_2(j/n_c) + a_3 + \varepsilon$, where a_1, a_2 and a_3 are coefficients of the corresponding plane, and ε is a zero-mean error term, usually assumed to be identically and independently distributed. It is assumed that pixels of the same region share the same a_1, a_2 and a_3 .

To generalise $\delta_O(x, R)$, first obtain least squares estimates of the coefficients using the pixels in R ; denote the resulting estimates as $\hat{a}_1, \hat{a}_2$ and $\hat{a}_3$. If the coordinate of pixel x is $(i_x/n_r, j_x/n_c)$, then the new homogeneity criterion is

$$\delta_L(x, R) = \left| g(x) - \left\{ \hat{a}_1 \left(\frac{i_x}{n_r} \right) + \hat{a}_2 \left(\frac{j_x}{n_c} \right) + \hat{a}_3 \right\} \right|$$

Thus at each time step the pair of (x, R) that minimises $\delta_L(x, R)$ is first sought, and the region R is grown by adding x into it. Note that the estimates $(\hat{a}_1, \hat{a}_2, \hat{a}_3)$ will need to be re-calculated whenever a new x is added to R . However, fast updating formulae exist so computationally it is fast. In sequel this new variant is termed linear SRG (Linear-SRG).

2.3 Quadratic SRG

Quadratic SRG (Quadratic-SRG) is similar to Linear-SRG. The only difference is that the regions are modelled by quadratic surfaces instead of linear planes. Specifically, the greyvalues of a region are assumed to follow $b_1(i/n_r)^2 + b_2(i/n_r)(j/n_c) + b_3(j/n_c)^2 + b_4(i/n_r) + b_5(j/n_c) + b_6 + \varepsilon$ with

$b_1, \dots, b_6$ being the coefficients of the surface. For any pair (x, R) , denote the least squares estimates of these coefficients as $\hat{b}_1, \dots, \hat{b}_6$. The corresponding homogeneity criterion is

$$\delta_Q(x, R) = \left| g(x) - \left\{ \hat{b}_1 \left(\frac{i_x}{n_r} \right)^2 + \hat{b}_2 \left(\frac{i_x j_x}{n_r n_c} \right) + \hat{b}_3 \left(\frac{j_x}{n_c} \right)^2 + \hat{b}_4 \left(\frac{i_x}{n_r} \right) + \hat{b}_5 \left(\frac{j_x}{n_c} \right) + \hat{b}_6 \right\} \right| \quad (1)$$

It is straightforward to see that both Linear-SRG and Quadratic-SRG generalise the original SRG, as both $\delta_L(x, R)$ and $\delta_Q(x, R)$ reduce to $\delta_O(x, R)$ when the appropriate coefficients are set to zero. In other words, both Linear-SRG and Quadratic-SRG can handle regions of constant greyvalues, as the least squares estimates of the corresponding coefficients will be close to zero.

2.4 Stabilised SRG

When growing a region, the original SRG does not impose any constraint or restriction on the shape or boundary of the region. When the signal-to-noise ratio is low, the final segmented boundaries could be very rough even though the true boundaries were smooth. In addition, the so-called leakage problem could occur. This leakage problem refers to the situation when the greyvalues of two adjacent objects are similar, and the growing region of one object breaks the true boundary and enters into the other object through a narrow spike. This subsection proposes a new SRG variant that encourages smoother boundaries and also aims to prevent the leakage problem. When comparing with previous attempts [e.g. 6, 8], one advantage of the current proposal is that it can be straightforwardly paired with any of the SRG variants described above. The new variant is termed stabilised SRG (Stabilised-SRG).

During the growth process, when determining which pixel x should be added to one of the growing regions,

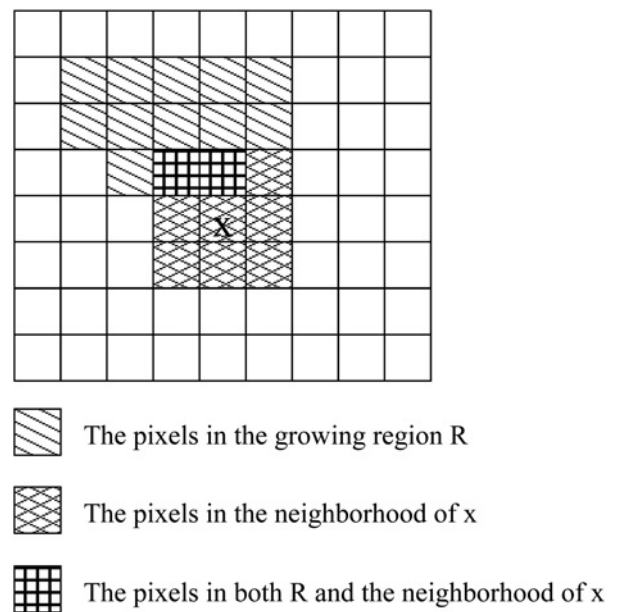


Fig. 1 Illustration of the set of neighbouring pixels Y_L (including x itself)

Stabilised-SRG does not only consider the greyvalue of x , but it also takes the greyvalues of a set of neighbouring pixels of x into account. By simultaneously considering the greyvalues of these neighbouring pixels, as to be shown below, will have a stabilising effect on how the regions should be grown. This set of neighbouring pixels are defined as all pixels covered by the square of size $(2L+1) \times (2L+1)$ centred at x ; the choice of L will be discussed below. In below this set of neighbouring pixels are denoted as $Y_L = \{y_{jk} | -L \leq j, k \leq L\}$. Note that $y_{00} = x$, but for convenience, y_{00} is used in Y_L instead of x . See Fig. 1 for an example with $L = 1$. With this setup, at each step Stabilised-SRG adds x to R if the following criterion is minimised (see (2)) where $\bar{r} = (1/n) \sum_{i=1}^n g(r_i)$ is the averaged greyvalue of R . From the definition of $\delta_S(x, R, Y_L)$, the stabilising effect on the growth is apparent: it favors adding pixel x to a growing region if the greyvalue of x is similar to its surrounding pixels. This property tends to produce smoother boundaries.

Note that when $L = 0$, $\delta_S(x, R, Y_L)$ reduces to $\delta_O(x, R)$, and therefore the original SRG is a special case of Stabilised-SRG. Also notice that Stabilised-SRG can be paired with Linear-SRG or Quadratic-SRG by modifying $\delta_S(x, R, Y_L)$, in a similar fashion as $\delta_O(x, R)$ was modified to $\delta_L(x, R)$ and $\delta_Q(x, R)$. The parameter L , which determines the size of the neighbourhood, also determines the extent to which the boundary will be smoothed. The larger the value of L , the smoother the boundary. The optimal choice of L will, amongst other factors, depend on the objects to be segmented. In practice it can be chosen by an operator in an interactive manner, or pre-specified if prior knowledge about the image is available.

2.5 Practical implementation

One attractive feature shared by all of the above SRG variants is that they are conceptually simple and their practical implementation does not involve any computationally expensive operations (e.g. function optimisation using iterative methods). In fact their practical implementation is similar to that of the original proposal of [1], which consists of the following steps:

Step 0: Construct the initial growing region sets $A_1, A_2, \dots, A_N$, each of which is composed of the pre-specified seeds. Label each seed pixel by the index of its corresponding set.

Step 1: For each set A_i , compute the value of the homogeneity criterion δ for all its immediate, unlabeled neighbours. The criterion δ can be any of δ_O , δ_L , δ_Q or δ_S .

Step 2: Take the pixel with the smallest value of δ from these immediate neighbours and add it to its corresponding growing region set. Label this pixel with the index of its corresponding set. If all the pixels in the image are labelled, the algorithm finishes. Otherwise, update all A_i 's and go to Step 1.

Finally it is worth mentioning that, depending on the nature of the image to be segmented, in practice one could achieve better results by mixing the above growing strategies; for example, grow some regions with Linear-SRG, while others with Stabilised-SRG. Section 3.3 below presents some empirical results for this mixing strategy.

2.6 Methods for evaluating segmentation results

There are two common types of methods for evaluating the results produced by a segmentation algorithm. They are supervised evaluation and unsupervised evaluation. Supervised evaluation is subjective in the sense that the quality of segmentation is evaluated by a human visually. In contrast, unsupervised evaluation is more objective and thus is more preferable under certain circumstances when the quality of segmentation is difficult to judge merely by eyeballs; for example, see [9].

For Stabilised-SRG, the evaluation of segmentation quality is based on whether leakage happens and/or whether boundaries are smooth, which can be examined straightforwardly by supervised evaluation. For the other two proposed variants, Linear-SRG and Quadratic-SRG, we recommend using the unsupervised evaluation method based on the so-called F 'evaluation function' ([10]). Suppose a digital image I has been segmented into N regions, denoted as $A_j, j = 1, 2, \dots, N$. For region j , denote its area (measured by number of pixels) as $S_j = |A_j|$. Suppose the true geometric shape of each region is either a linear plane or a quadratic surface. By fitting the corresponding parametric form to the greyvalues of the pixels in region j , we obtain the fitted greyvalues $\hat{g}(x)$, where $x \in A_j$. The generalised F 'evaluation function' is defined as

$$F_g(I) = \sqrt{N} \sum_{j=1}^N \frac{e_j^2}{\sqrt{S_j}}$$

where $e_j^2 = \sum_{x \in A_j} (g(x) - \hat{g}(x))^2$. Smaller F_g values indicate better segmentation results.

3 Empirical results

In this section the proposed SRG variants are applied to different simulated and real images to illustrate their empirical performances.

3.1 Simulated images

Three artificial images, all of size 100×100 , are used. The greyvalue $I_1(i, j)$ of the first image at the i th row and j th

$$\begin{aligned} \delta_S(x, R, Y_L) &= \frac{1}{(2L+1)^2} \left\{ \sum_{j,k=-L, \dots, L; j^2+k^2 \neq 0} \left| g(y_{jk}) - \frac{1}{n} \sum_{i=1}^n g(r_i) \right| + \left| g(x) - \frac{1}{n} \sum_{i=1}^n g(r_i) \right| \right\} \\ &= \frac{1}{(2L+1)^2} \left\{ \sum_{j,k=-L}^L \left| g(y_{jk}) - \frac{1}{n} \sum_{i=1}^n g(r_i) \right| \right\} \\ &= \frac{1}{(2L+1)^2} \left\{ \sum_{j,k=-L}^L \left| g(y_{jk}) - \bar{r} \right| \right\} \end{aligned} \quad (2)$$

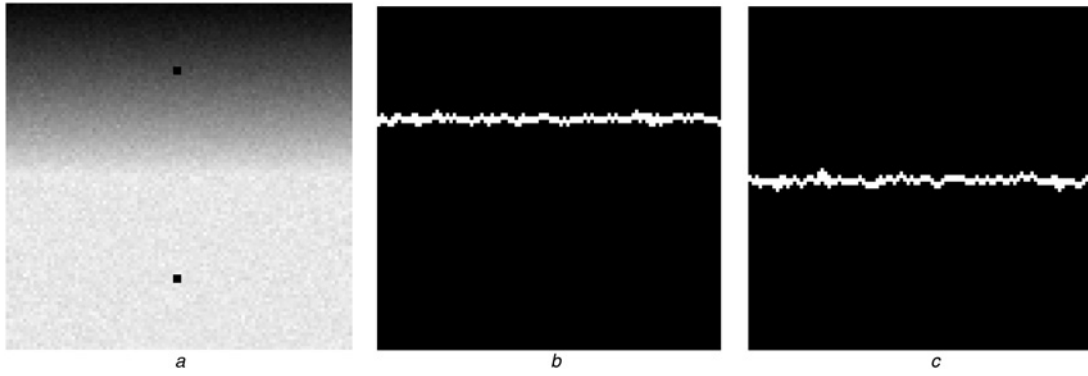


Fig. 2 Comparison of segmentation results from original SRG and Linear-SRG for image $I_1(i, j)$

- a* Image $I_1(i, j)$ with two seeds shown as black squares
b Segmentation result from original SRG
c Segmentation result from Linear-SRG

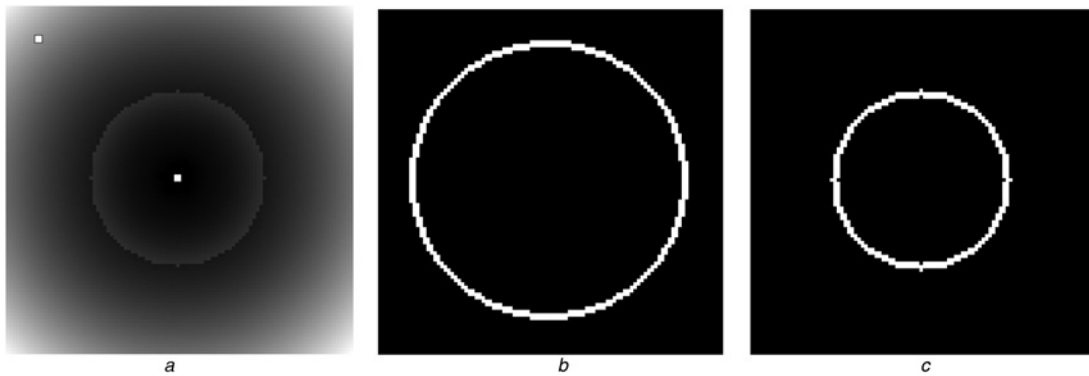


Fig. 3 Comparison of segmentation results from original SRG and Quadratic-SRG for image $I_2(i, j)$

- a* Image $I_2(i, j)$ with two seeds shown as white squares
b Segmentation result from original SRG
c Segmentation result from Quadratic-SRG

column is given by

$$I_1(i, j) = \begin{cases} 2(i - 50) + 10 + \epsilon, & i \in [1, 50] \\ 10 + \epsilon, & i \in [51, 100] \end{cases}$$

where the error term ϵ is normally distributed with mean 0 and standard deviation $\sigma=2$. This is a noisy image consisting of two neighbouring planar surfaces with boundary at $i=50$. The first plane is with a non-zero slope while the second plane is horizontal. Both the original SRG and the proposed Linear-SRG were applied to segment this image. The results are given in Fig. 2. One can see that Linear-SRG outperformed the original SRG as its segmented boundary is much closer to $i=50$.

The second artificial image $I_2(i, j)$ is composed of two quadratic surfaces, both centred at $(50, 50)$ (see (3))

where the noise standard deviation is $\sigma=2$. Fig. 3 indicates that Quadratic-SRG gave superior results to the original SRG, as it nearly recovered the true boundary defined by the circle $(i - 50)^2 + (j - 50)^2 = 25^2$.

The last artificial image $I_3(i, j)$ is similar to that in [1]

$$I_3(i, j) = \begin{cases} -10 + \epsilon, & i \in [1, 40] \\ i - 51 + \epsilon, & i \in [41, 60] \\ 10 + \epsilon, & i \in [61, 100] \end{cases}$$

where $\sigma=10$, a relatively high level of noise. Although there are three segments in $I_3(i, j)$, it is typically regarded as an image with two planes, with a blurred boundary at $i=50$. Both the original SRG and Stabilised-SRG with $L=10$ are applied and the results are given in Fig. 4. It can be seen that the boundary produced by Stabilised-SRG is smoother.

3.2 Real images

In this and the next subsections real images are used to test the performances of the new SRG variants. These real images have been used by other authors as testing images; see USF Range Image Database, SAR processing tools plug-in, [11–14]. In the sequel, we will focus on comparing the results obtained from the original SRG and those from the proposed variants of SRG.

$$I_2(i, j) = \begin{cases} (i - 50)^2 + (j - 50)^2 + \epsilon, & (i - 50)^2 + (j - 50)^2 \leq 25^2 \\ 0.5\{(i - 50)^2 + (j - 50)^2\} + \epsilon, & \text{otherwise} \end{cases} \quad (3)$$

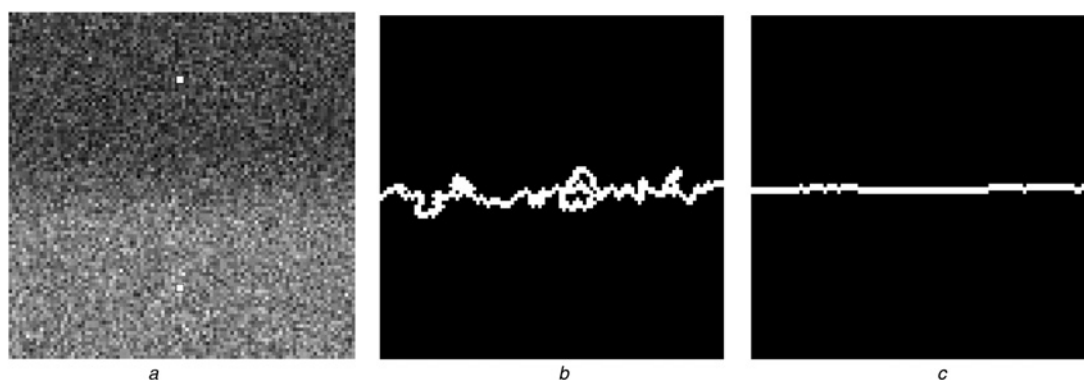


Fig. 4 Comparison of segmentation results from original SRG and stabilised-SRG for image $I_3(i, j)$

a Image $I_3(i, j)$ with two seeds shown as white squares

b Segmentation result from original SRG

c Segmentation result from Stabilised-SRG with $L = 10$. Note that the 'loops' in (b) are actually not loops, they are artefacts caused by how the boundary is displayed

Regions in the first image are polyhedron faces and they are ideal for testing range image segmentation methods. The second image is a scenery of a beach and it would be useful to separate the sky, sand and water. Both the original SRG and Linear-SRG are applied to these images, and the results are shown in the top two rows of Fig. 5. Given the fact that those regions inside these images are better modelled with linear planes, it is not surprising to see Linear-SRG outperformed the original SRG.

The third image contains a number of potatoes and quadratic modelling seems to be appropriate. The last row of Fig. 5 shows the result of Quadratic-SRG, which shows a substantial improvement over the original SRG.

The fourth and fifth images under test are two synthetic aperture radar (SAR) images of a river outlet and a rural area respectively. As with other SAR images, they are relatively noisy which would lead to rough segmented boundaries and leakages. The segmentation results from

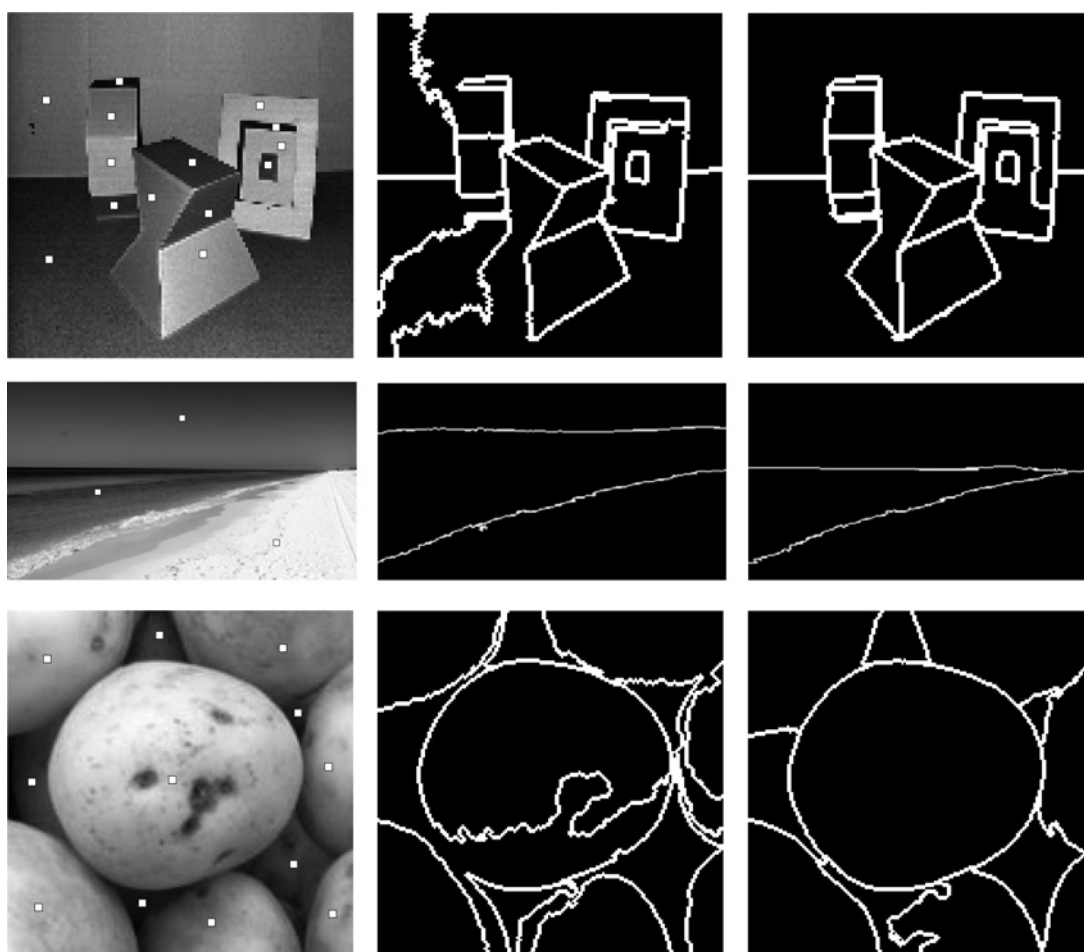


Fig. 5 Comparison of segmentation results from original SRG, Linear-SRG and Quadratic-SRG for real images

Left column: real images with seeds shown as white squares. Middle column: segmentation results from original SRG. Right column, top two rows: segmentation results from Linear-SRG. Right column, last row: segmentation results from Quadratic-SRG

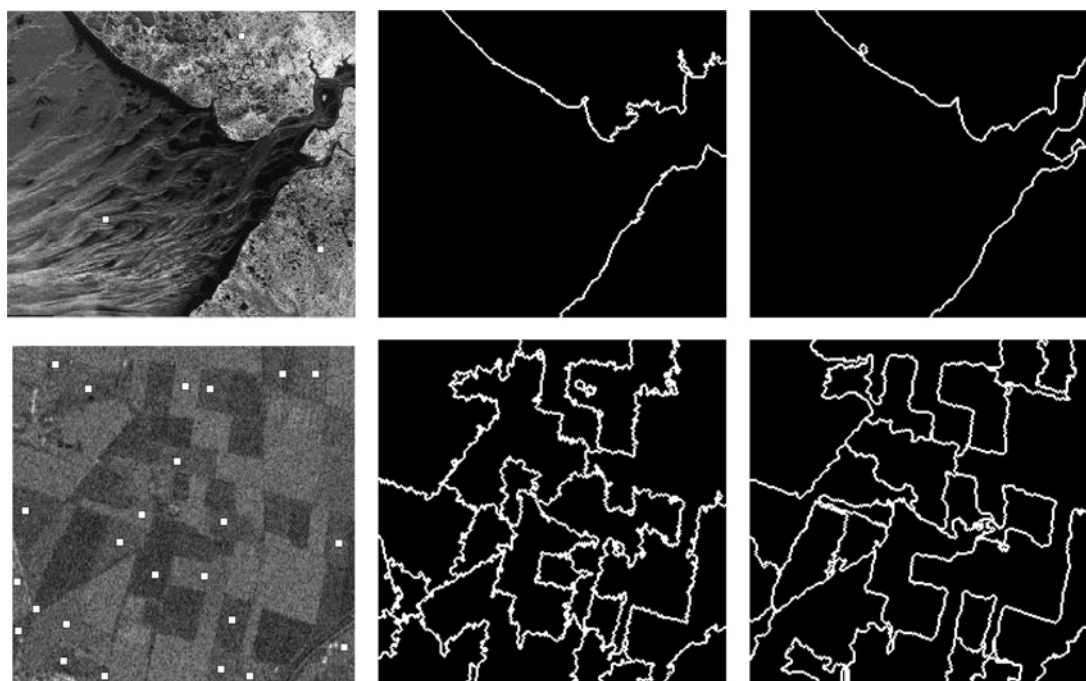


Fig. 6 Comparison of segmentation results from original SRG and Stabilised-SRG for real images

Left column: real images with seeds shown as white squares. Middle column: segmentation results from original SRG. Right column: segmentation results from Stabilised-SRG with $L=2$

Stabilised-SRG with $L=2$ are given in Fig. 6. When comparing with results from the original SRG, Stabilised-SRG segmented the regions more accurately and with smoother boundaries.

3.3 Mixing the variants

This final set of numerical experiments illustrate the possibility of mixing different growing strategies together. In the first two

images displayed in Fig. 7, one can see that some regions can be well approximated by horizontal planes while others by quadratic surfaces. Therefore it is natural to grow some of the regions using the original SRG, while others with Quadratic-SRG. The results are also shown in Fig. 7, which suggest that the mixing strategy is superior for these two images when comparing with the original SRG.

Finally we mix the original SRG with Stabilised-SRG. The two images under consideration are an X-ray image of a

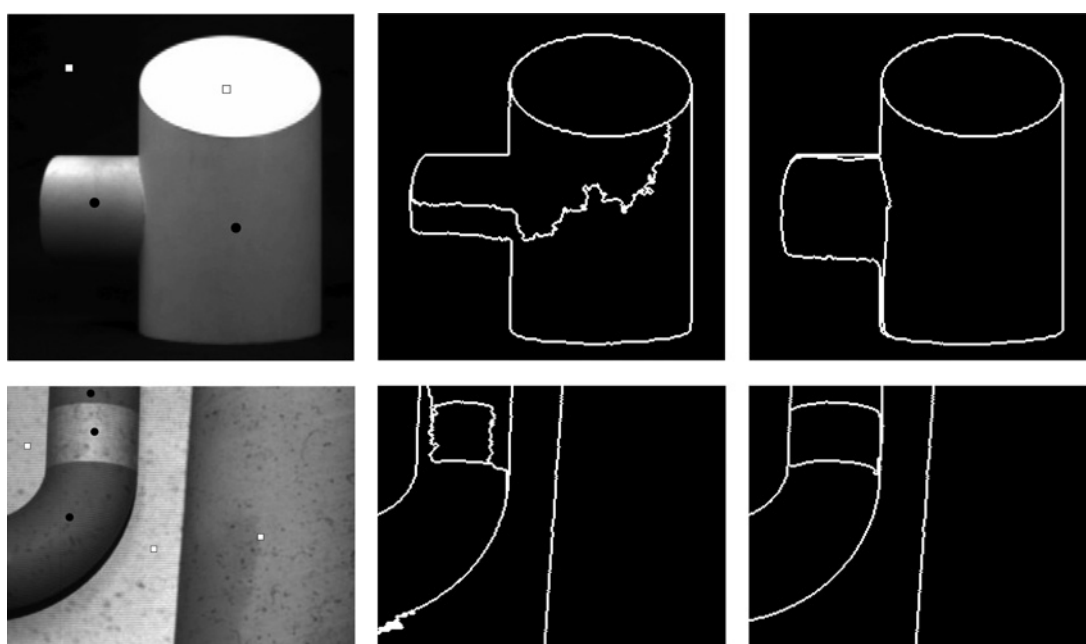


Fig. 7 Comparison of segmentation results from original SRG and it mixed with Quadratic-SRG for real images

Left column: real images of cylinders and pipelines with seeds shown as white squares and black circles. Middle column: segmentation results when all seeds were grown using original SRG. Right column: segmentation results when the white and black seeds were grown, respectively, using original SRG and Quadratic-SRG

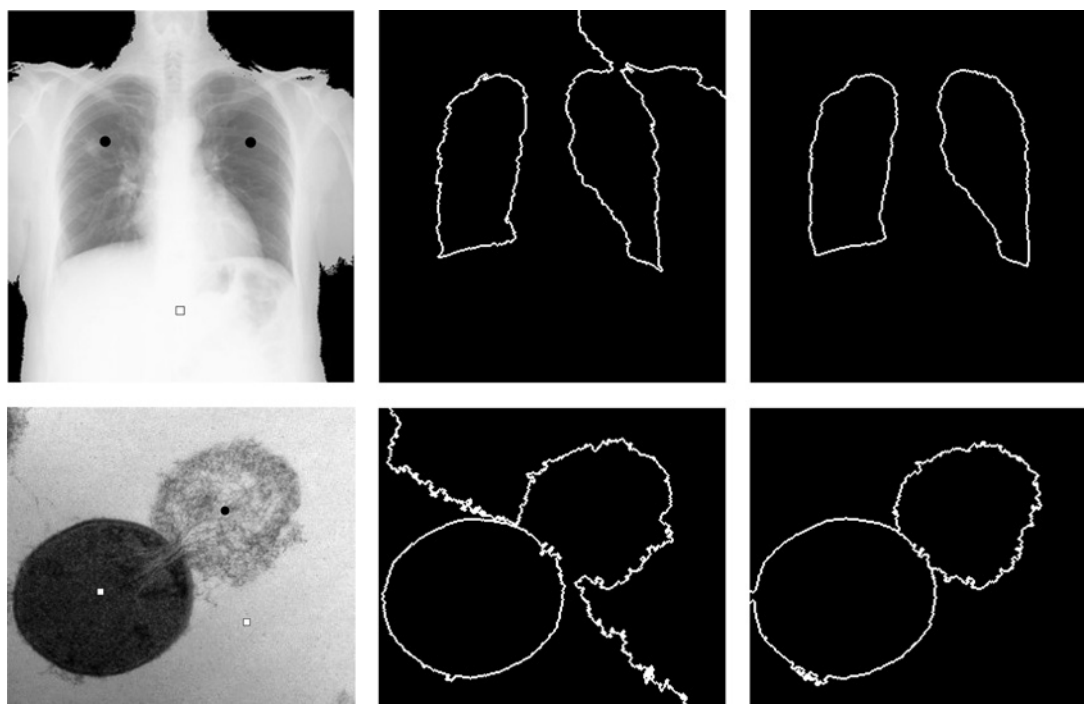


Fig. 8 Comparison of segmentation results from original SRG and it mixed with Stabilised-SRG for real images

Left column: real images of a human lung and biological cells with seeds shown as white squares and black circles. Middle column: segmentation results when all seeds were grown using original SRG. Right column: segmentation results when the white and black seeds were grown, respectively, using original SRG and Stabilised-SRG with $L = 10$ for the first image and $L = 20$ for the second one

Table 1 F_g values for segmented images produced by the original SRG and variants of SRG

Figure index	Original SRG	Variants of SRG
Fig. 2	50 586	4230.6
Fig. 3	6743.9	7.575
first row of Fig. 5	264 900	207 970
second row of Fig. 5	196 880	125 050
third row of Fig. 5	367 130	312 330
first row of Fig. 7	174 460	86 007
second row of Fig. 7	112 170	101 170

human lung and a microscope image of two biological cells; see Fig. 8, which also shows the segmentation results. From these one can see that the leakage problems presented in the original SRG results are rectified by mixing with Stabilised-SRG.

3.4 Segmentation evaluation given by F_g

Table 1 lists the F_g values for different segmented images produced by the original SRG and variants of SRG (including Linear-SRG and Quadratic-SRG). It can be seen that the variants of SRG outperformed the original SRG because the former always achieved smaller F_g values. This empirical conclusion is consistent with the previous supervised evaluation obtained visually.

4 Concluding remarks

In this paper three new variants of SRG are presented: Linear-SRG, Quadratic-SRG and Stabilised-SRG. These new variants are conceptually simple and improve the performance of the original SRG under different

circumstances. When the greyvalues of the regions can be well modelled by linear planes or quadratic surfaces, it is shown that Linear-SRG and Quadratic-SRG outperform the original SRG, while Stabilised-SRG tends to produce smoother boundaries than the original SRG in noisy images.

These variants can be further generalised. For example, they can be straightforwardly extended to multiband images (e.g. colour images). For Linear-SRG and Quadratic-SRG, they can also be extended to higher order modelling, although numerical instability may pose a problem if the order is too high. For Stabilised-SRG, one could modify the shape of the neighbouring set Y_L in (2); currently Y_L is a square. For example, one could use a slanted elongated shape (e.g. a rectangle or a line) to encourage stabilisation in a certain direction.

Caution should be exercised when applying the proposed variants of SRG in practice. For example, one should be careful if the underlying assumptions of these SRG variants are violated. That is, when applying Linear-SRG or Quadratic-SRG, linear planes or quadratic surfaces should be good approximations for the regions. Otherwise, more complicated geometric shapes of region surfaces should be considered. Another example for Stabilised-SRG is when the boundary between a region and the background is very blurry. In this case, even though Stabilised-SRG has the effect of leakage prevention, the leakage may still happen because of the extremely ambiguous boundary. One possible solution to overcome this issue is to incorporate prior knowledge of the shape of the region into the learning algorithm.

Additionally, the proposed algorithms may fail when there is no sufficient number of pixels in a region. Fitting such a region with a linear plane or a quadratic surface would be unstable and inaccurate. In this case, one should rather

apply original SRG instead of Linear-SRG or Quadratic-SRG since taking average is more robust to the limited number of pixels.

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